

CHERNUSHKA, Russian Federation

WMO: 284280

Lat: 56.500N

Lon: 56.133E

Elev: 148

StdP: 99.56

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-28.7	-35.0	0.1	-32.0	-31.4	0.2	-28.6	10.1	-6.5	9.1	-6.4	1.3	180	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.2	20.4	28.3	19.5	26.5	18.7	21.6	28.2	20.6	26.6	19.6	25.0	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.3	24.5	18.4	13.5	23.6	17.5	12.8	22.6	63.4	28.3	60.0	27.0	56.6	24.8	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.9	7.8	6.9		DB	-37.2	32.3	3.7	2.2	-39.8	33.9	-42.0	35.2	-44.1	36.5	-46.7	38.1
				WB	-37.3	23.1	3.7	1.0	-39.9	23.8	-42.1	24.4	-44.2	24.9	-46.8	25.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-13.9	-12.8	-5.2	3.6	11.8	16.3	18.5	16.4	10.7	3.3	-4.0	-11.9	(6)
	DBStd	12.84	7.88	7.46	5.06	5.25	4.69	4.27	3.83	4.13	4.20	4.41	5.89	7.98	(7)
	HDD10.0	3443	739	638	471	202	36	5	0	2	39	212	421	678	(8)
	HDD18.3	5772	998	872	729	443	207	88	46	87	231	466	671	936	(9)
	CDD10.0	826	0	0	0	10	92	193	264	201	61	4	0	0	(10)
	CDD18.3	112	0	0	0	0	4	26	52	28	2	0	0	0	(11)
	CDH23.3	1313	0	0	0	4	97	302	576	308	26	0	0	0	(12)
	CDH26.7	368	0	0	0	0	13	78	177	96	4	0	0	0	(13)
(14) Wind	WSAvg	3.0	3.2	3.3	3.5	3.1	3.1	2.7	2.2	2.3	2.6	3.3	3.4	3.2	(14)
(15) Precipitation	PrecAvg	588	38	28	31	27	44	72	70	63	58	64	51	41	(15)
	PrecMax	767	78	57	56	66	77	150	153	131	152	111	84	68	(16)
	PrecMin	390	7	10	3	1	17	17	15	15	18	16	11	13	(17)
	PrecStd	82	16	12	15	18	19	34	39	29	30	25	22	16	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.6	7.5	22.9	28.3	31.9	32.7	32.5	26.8	18.2	9.4	1.6	(19)
		MCWB	0.4	0.2	3.2	13.4	16.8	20.8	21.0	20.4	16.0	11.1	7.7	0.8	(20)
	2%	DB	-0.2	0.6	4.7	18.6	26.2	28.7	30.4	29.4	22.7	14.7	6.5	0.9	(21)
		MCWB	-0.7	-0.1	1.8	11.4	15.9	19.5	20.8	20.1	15.5	9.3	5.6	0.4	(22)
	5%	DB	-1.3	-0.5	2.9	15.0	23.8	26.7	28.6	26.7	20.0	12.0	4.2	0.0	(23)
		MCWB	-1.8	-1.2	0.7	8.8	14.8	19.0	20.3	19.0	14.5	8.4	3.3	-0.6	(24)
	10%	DB	-3.1	-2.4	1.5	11.7	21.4	24.7	26.7	24.2	17.7	9.9	2.2	-1.4	(25)
		MCWB	-3.7	-3.3	0.0	7.2	13.8	17.9	19.6	18.0	13.2	7.9	1.4	-2.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	0.8	4.0	13.9	18.8	22.1	22.8	21.8	17.9	12.0	8.0	1.2	(27)
		MCDB	1.0	1.2	6.7	21.9	25.7	29.3	29.1	29.4	22.7	16.5	9.3	1.3	(28)
	2%	WB	-0.6	-0.1	2.4	11.8	17.0	20.8	21.9	20.7	16.2	10.2	5.5	0.5	(29)
		MCDB	-0.3	0.6	4.1	17.6	23.8	26.8	28.4	27.4	20.9	13.1	6.3	0.8	(30)
	5%	WB	-1.8	-1.2	1.2	9.6	15.6	19.7	21.1	19.6	14.9	9.0	3.4	-0.5	(31)
		MCDB	-1.4	-0.4	2.4	14.2	22.1	25.2	27.2	25.3	19.4	11.1	4.0	0.0	(32)
	10%	WB	-3.7	-3.2	0.2	7.5	14.3	18.5	20.1	18.5	13.8	7.9	1.6	-2.1	(33)
		MCDB	-3.1	-2.5	1.4	11.4	20.6	23.6	25.5	23.5	17.2	9.9	2.1	-1.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.7	8.9	9.4	11.8	11.2	11.2	10.6	9.4	6.0	4.8	6.3	(35)
		MCDBR	5.3	5.7	9.1	14.3	16.3	14.6	14.5	15.0	13.4	11.6	4.8	4.1	(36)
		MCWBR	5.3	5.3	6.8	8.0	7.5	6.8	6.2	6.5	6.8	7.1	4.2	4.2	(37)
	5% WB	MCDBR	5.2	5.4	6.8	13.3	14.3	12.9	12.9	13.6	11.6	10.3	4.7	4.5	(38)
		MCWBR	5.3	5.1	5.3	7.7	7.5	7.1	6.1	6.5	6.3	7.1	4.4	4.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.296	0.320	0.375	0.409	0.398	0.393	0.414	0.435	0.381	0.355	0.315	0.293	(40)
	taud		2.007	2.012	1.923	2.053	2.265	2.336	2.286	2.203	2.323	2.257	2.113	1.978	(41)
	Ebn at Noon		542	700	750	797	839	849	819	768	762	669	524	429	(42)
	Edh at Noon		65	98	139	143	123	116	120	123	94	77	59	52	(43)
(44) All-Sky Solar Radiation	RadAvg		0.66	1.61	2.98	4.23	5.24	5.55	5.58	4.26	2.61	1.14	0.53	0.38	(44)
	RadStd		0.09	0.23	0.30	0.51	0.56	0.53	0.60	0.41	0.24	0.23	0.11	0.09	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page